MIL-C-915/49B
5 April 1973
SUPERSEDING[1]
MIL-C-915/49A
1 August 1972

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL TYPE 2SWU

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the cable described herein shall consist of this document and the latest issue of Specification MIL-C-915.

REQUIREMENTS:

Qualification required.

Construction (Watertight)

(B)

First - Copper conductor, AWG No. 18 (7 X 0.0152 inch) tin coated. Maximum diameter 0.050 inch.
Second - Polyethylene insulation, type II, class L, grade 4 of L-P-390. Nominal thickness 0.020 inch. Colored insulation, one black and one white conductor for each pair.
Third - Clear polyamide jacket on each conductor, type III, grade E or type IV of MIL-M-20693. Nominal thickness 0.003 inch.
Fourth - Two conductors (one black and one white) cabled together with a nominal lay of 3 inches to form each pair. For 2SWU-1[2].
Fifth - Binder tape over each pair, at manufacturer's option, applied helically with overlap.
Sixth - Braided shield of AWG No. 34 or No. 36 tin coated copper. Braid angle 30 to 35 degrees; minimum coverage of 85 percent.
Seventh- Shield insulation of one polyester tape, type G of MIL-I-631, plus a jacket of clear polyamide, type III, grade E or type IV of MIL-M-20693. Nominal thickness of 0.003 inch. Standard Identification Code, applied by Method 2.
- Alternate shield insulation of two polyester tapes, type G of MIL-I-631, sealed. Standard Identification Code, applied by Method 2, on inner tape.
- Diameter over completed pair (see details).
Eighth - The required number of pairs (see details) cabled with a lay not greater than 24 times the pitch diameter of the layer. Cabling sequence shall be consecutive, starting with pair No. 1, from center outward. Non-fibrous filler shall be employed to form a firm round cable.
Ninth - Binder tape applied helically with overlap.
Tenth - Standard thermoplastic jacket (gray). (See details for thickness.) Cable surface marking required.

Details

Type and size	Number of pairs	Diameter over completed pair nominal (inch)	Cable jacket thickness min. (inch)	Overall diameter	
				Min. (inches)	Max. (inches)
2SWU-1	1		0.012	0.240	0.255
2SWU-3	3	0.213	.060	.670	.710
2SWU-7	7	.213	.060	.860	.910
2SWU-12	12	.213	.070	1.130	1.200
2SWU-19	19	.213	.080	1.292	1.380

[1] Requirements formerly covered on MIL-C-24145/9.
[2] For 2SWU-1, omit seventh, eighth and ninth.

(B) Denotes changes.

FSC 6145

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T12:11Z
Check the source to verify that this is the current version before use.

Details (Cont'd)

Type and size	Number of pairs	Diameter over completed pair nominal (inch)	Cable jacket thickness min. (inch)	Overall diameter	
				Min. (inches)	Max. (inches)
2SWU-24	24	0.213	0.080	1.500	1.590
2SWU-30	30	.213	.080	1.670	1.760
2SWU-37	37	.213	.100	1.785	1.870
2SWU-61	61	.213	.100	2.205	2.300

EXAMINATION AND TESTS:

	Requirements:
Basic electrical:	
(B) Conductor resistance - At 25°C., ohms/1000 ft., maximum	7.35
Voltage withstand - Volts, rms, minimum	
Conductor to conductor	2000
Conductor to shield	1000
Shield to shield	500
Shield to water	500
Group A:	
Visual and dimensional	
Hydrostatic (open end) - Leakage at 300 psi, 2 hrs., cu. in., maximum	1.0
Capacitance	
Mutual capacitance - At 1 kHz, pF/ft., maximum	30
Capacitance unbalance - Percent, maximum	10
Group B:	
Drip - At 75° ± 1°C.,	Zero
Physicals (unaged)	
Insulation (conductor)	
Tensile strength - psi, minimum	1400
Elongation - Percent, minimum	300
Jacket (cable)	
Tensile strength - psi, minimum	1800
Elongation - Percent, minimum	225
Group C:	
Physicals (aged)	
Insulation (conductor)	
Air oven	
Elongation - Percent of unaged, minimum.	60
Jacket (cable)	
Hot oil immersion	
Tensile strength - psi, minimum	1400
Elongation - Percent, minimum	50
Permanence of printing (jacket) - Cycles, minimum.	250
Cable filler removability (all types, except 2SWU-1)	

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical plus all of Groups A, B and C plus the following:	
Cable aging and compatibility (95° ± 3°C.) (all types, except 2SWU-1)	No failure

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T12:11Z
Check the source to verify that this is the current version before use.

UNIT ORDERING LENGTHS:

Size	Feet (nominal)
1 through 12	1500
19 through 24	1000
30 through 61	500

Custodians:
Army - MI
Navy - SH
Air Force - 80
Review activities:
Army - EL, MI, AV
Navy - EC
Air Force - 80
User activities:
Army - ME, MU, WC
Navy - CG

Preparing activity:
Navy - SH
(Project 6145-0648-9)

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T12:11Z
Check the source to verify that this is the current version before use.